

Sine function

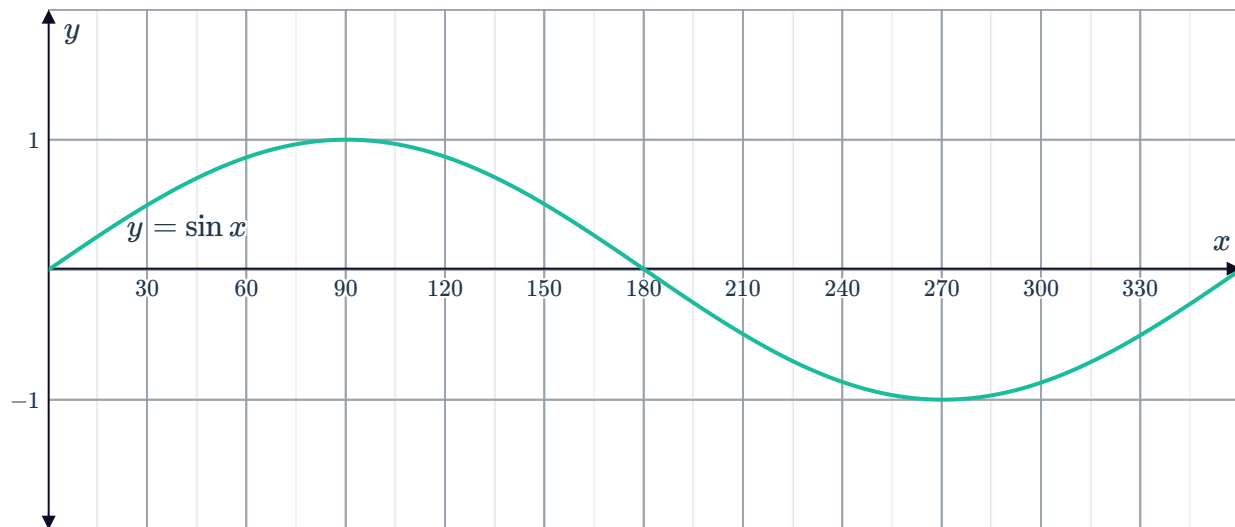


1

a

x	0°	30°	90°	150°	180°	210°	270°	330°	360°
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b



c Negative

d Quadrant 4

e The function values and shape of the graph repeat at regular intervals.

f Period: 360°

g (0,0)

h 1

i -1

j $[-1,1]$

2



a

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0

b

c Period: 2π

3

a Because the range of the unit circle is $-1 \leq y \leq 1$.b Because the unit circle repeats every 360° .

4

a Yes

b No

c Yes

d No

e Yes

f No

g Yes

5

a

i Decreasing

ii Increasing

iii Decreasing

iv Increasing

b

i 180° ii -180°

6



a

- i Decreasing ii Increasing iii Decreasing iv Increasing

b

- i π ii $-\pi$

7

a 50 000

b The peak number of visitors to Times Square each day is 50 000.

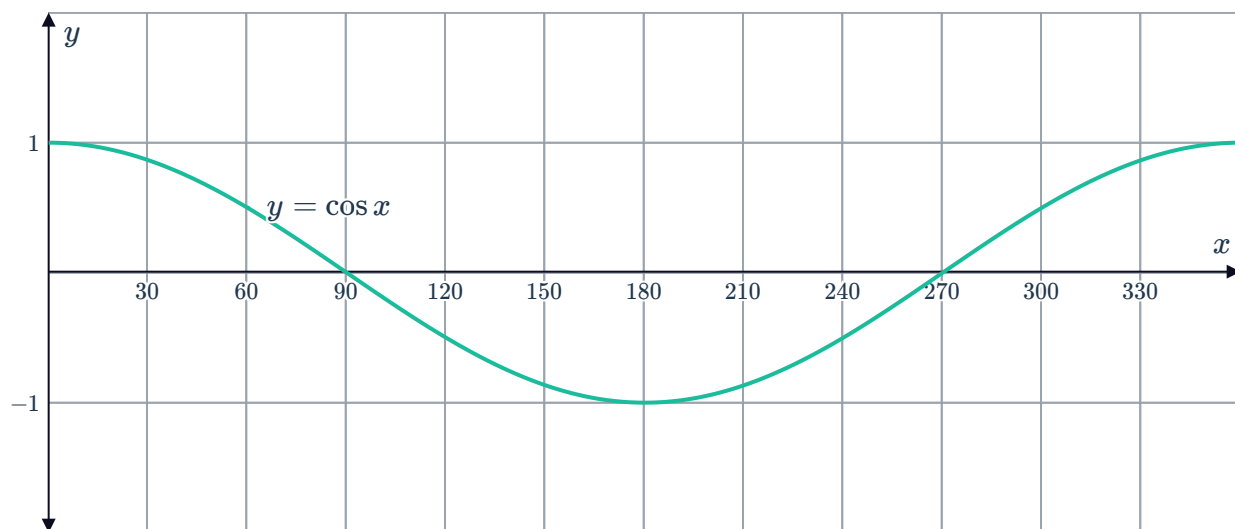
Cosine function

8

a

x	0°	60°	90°	120°	180°	240°	270°	300°	360°
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c Positive

d Quadrant 4

e (0, 1)

f 1

g -1

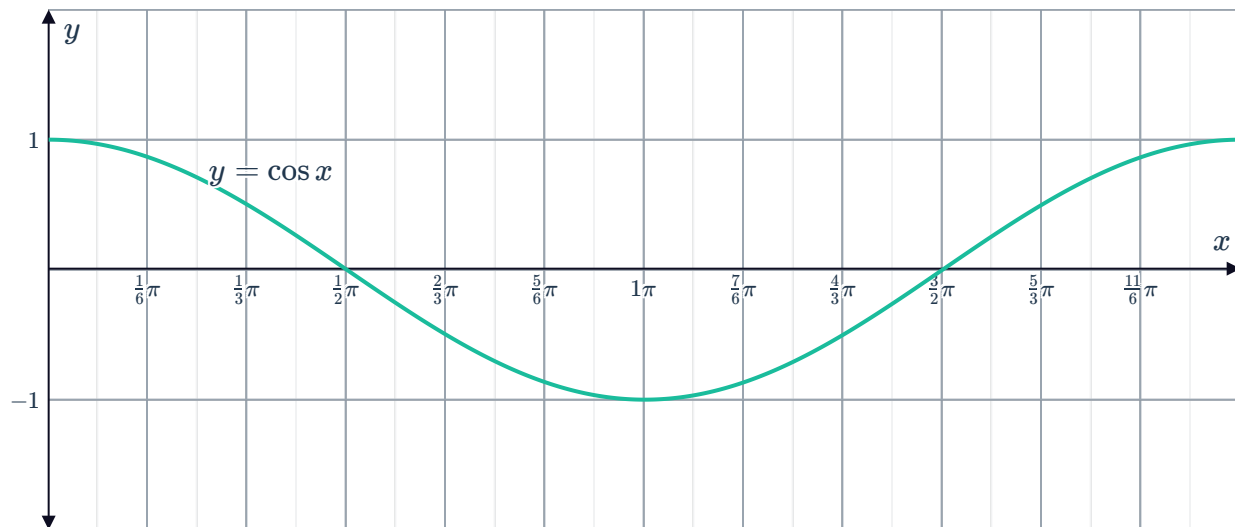
h $-\infty < x < \infty$

9 a



x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



10

- a Because the domain of the unit circle is $-1 \leq y \leq 1$.
b Because the unit circle repeats every 360° .

11

- a Yes b No c No d Yes
e No f No g Yes

12

- a 270°
b
i Increasing ii Decreasing iii Decreasing iv Increasing
c
i $90^\circ, 270^\circ$ ii $-270^\circ, -90^\circ$

13



a $\frac{3\pi}{2}$

b

i Increasing

ii Decreasing

iii Decreasing

iv Increasing

c

i $\frac{\pi}{2}, \frac{3\pi}{2}$

ii $-\frac{3\pi}{2}, -\frac{\pi}{2}$

14

a No

b Yes

c No

d No

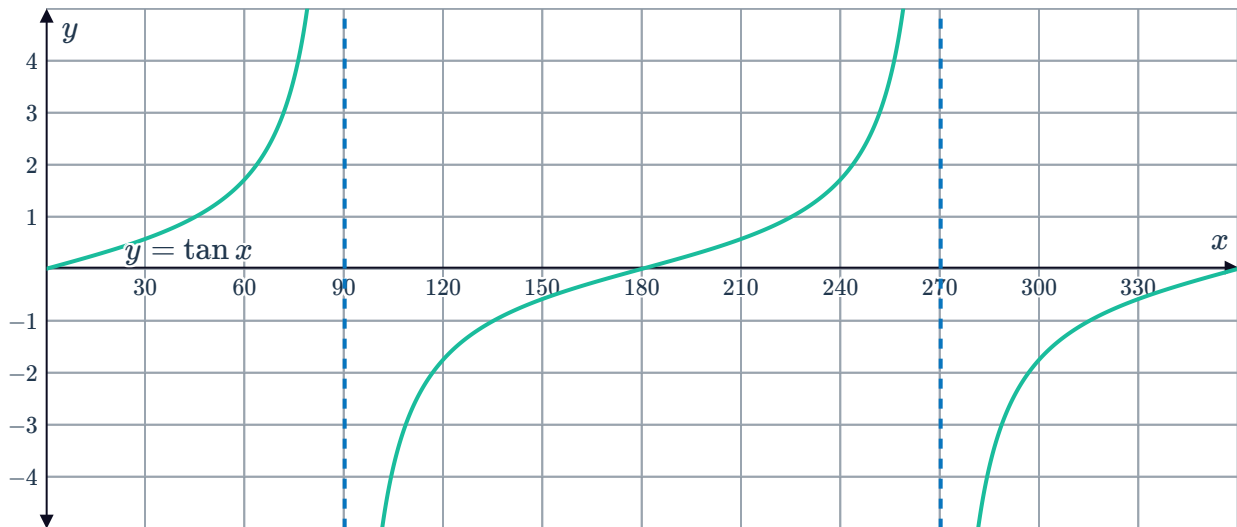
Tangent function

15

a

x	0°	60°	90°	120°	135°	180°	225°	270°	300°	360°
$\tan x$	0	$\sqrt{3}$	—	$-\sqrt{3}$	-1	0	1	—	$-\sqrt{3}$	0

b



c (0,0)

d Negative

e Quadrant 4

16

a Yes

b No

c No

d Yes

17 a 180°

b $-\infty < y < \infty$



630°

18

a $y = 0$

b π

c $x = \frac{\pi}{2}, \frac{3\pi}{2}$

d Increase

e $x = \pi$

19

a π

b $-2\pi, -\pi, 0, \pi, 2\pi$

c $\frac{\pi}{4}$

d $\frac{5\pi}{4}, -\frac{3\pi}{4}, -\frac{7\pi}{4}$

e $\frac{3\pi}{4}, \frac{7\pi}{4}, \frac{-\pi}{4}, \frac{-5\pi}{4}$

20

a The opposite side increases.

b The value of $\tan \theta$ increases.

c $\tan \theta = \frac{\sin \theta}{\cos \theta}$

d $\theta = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

e $\theta = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

f

θ	-2π	$-\frac{7\pi}{4}$	$-\frac{5\pi}{4}$	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{4}$
$\tan \theta$	0	1	-1	0	1	-1

θ	0	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{7\pi}{4}$	2π
$\tan \theta$	0	1	-1	0	1	-1	0

g

